

Amazon EC2 ▾

Amazon EC2 M8a instances

New general-purpose instances powered by 5th Generation AMD EPYC processors



Amazon Elastic Compute Cloud (Amazon EC2) M8a and M8azn instances are next generation general purpose instances powered by 5th Gen AMD EPYC processors. M8a instances deliver up to 30% better performance than M7a instances and are ideal for applications that benefit from high performance and high throughput. M8azn instances offer the highest maximum CPU frequency in the cloud at 5 GHz and deliver up to 2x higher compute performance compared to M5zn instances, with up to 4.3x higher memory bandwidth and 10x larger L3 cache, making them optimized for computationally and network-intensive workloads.

M8a instances are ideal for general purpose applications such as gaming, financial applications, ML inferencing, video encoding, application servers, mid-size data stores, application development environments, and caching fleets. M8azn instances are ideal for computationally and network-intensive applications such as real-time financial analytics, high-performance computing, high-frequency trading (HFT), CI/CD, intensive gaming, and simulation modeling for the automotive, aerospace, energy, and telecommunication industries.



Benefits

Accelerate performance

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Scale with flexibility

Features

Powered by 5th Generation AMD EPYC processors



High-performance interface

Supports diverse workloads

Built on AWS Nitro System

Product Details

Amazon EC2 M8a instances are powered by 5th Generation AMD EPYC processors and are an ideal fit for general-purpose workloads.

Instance size

m8a.medium

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m8a.xlarge

m8a.2xlarge

m8a.4xlarge

m8a.8xlarge

m8a.12xlarge

m8a.16xlarge



M8azn

Amazon EC2 M8azn instances are powered by 5th Generation AMD EPYC processors with the highest maximum CPU frequency in the cloud at 5 GHz and are an ideal fit for computationally and network-intensive workloads.

Instance size

m8azn.medium

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m8azn.xlarge

m8azn.3xlarge

m8azn.6xlarge

m8azn.12xlarge

m8azn.24xlarge

m8azn.metal-12xl



Customers



BLOCKTECH

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BlockTech

"BlockTech is a crypto-native trading firm that combines innovation, speed, and precision to stay at the cutting edge of algorithmic derivatives and spot trading. With Amazon EC2 M8azn instances, we observed significant performance gains over M8a. In our testing, CPU task processing improved by up to 20%, with packet processing performance improving by up to 30%, driven by the higher CPU frequency and network-optimized design. Under heavy load, M8azn delivered more consistent performance, enabling us to handle more trading volume during periods of peak demand."

Sander van Schie, Cloud Engineer, BlockTech



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Capital One



"The 7th-generation of EC2 instances with AMD processors provided a step-function increase in performance for the cost. With a 43% performance gain demonstrated in our benchmarking, we see the upcoming M8a instances as a continuation of this essential trend for our cloud infrastructure."

Brent Segner, Senior Distinguished Engineer | Senior Director, Capital One

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Fatshark



"At Fatshark, we're always looking for new tools that let us create higher intensity, challenging gameplay features for our players. Due to our signature horde gameplay, cloud VM CPU frequency and cache tend to be a constraint for us. During testing with M8azn instances, we saw 20% lower CPU utilization and 35% faster p90 frame-times for our most intense, high difficulty game modes compared to C7a instances. M8azn's 5GHz frequency, larger L3 cache, and higher memory bandwidth give us the headroom to innovate with our game modes and still deliver a more responsive experience to players."

Andrew Claridge, Technical Director, Fatshark

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Half Moon Bay Trading



"Amazon EC2 M8azn instances have provided significant performance improvements for our high frequency trading workloads. In our testing, with M8azn instances, we have observed a 13.7% improvement in median tick-to-trade latencies and a 20.1% improvement in tail latencies compared to M8a instances. Moreover, the 5GHz turbo frequency and high networking bandwidth offered by these instances has provided a meaningful speed-up in our applications that are dependent on CPU frequency and networking capabilities, allowing us to maintain high-speed execution during high-volume market events, directly contributing to our competitive positioning."

Zijun Hui, Chief Technology Officer, Half Moon Bay Trading

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**WOLFRAM**

Wolfram Research



"With the new M8azn instances, our Wolfram Language benchmarks show 69% improvement in numeric and symbolic computation performance compared to M7a. M8azn's 5GHz CPU frequency, larger L3 cache, and higher memory bandwidth enable faster complex computations, experiments, and simulations. Combined with Wolfram Language's integrated algorithms and built-in AWS connectivity, M8azn instances accelerate academic, R&D, and production applications."

Jesse Friedman, Software Engineer, Wolfram Research

Next steps

Start using Amazon EC2 M8a instances

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